# Junghun Hwang (황정헌)

stevehwang1210@gmail.com | **Portfolio: steveandy-sudo.github.io**

## Education

### Konkuk University

Mar 2022 - Expected Feb 2028
B.S. in Smart Vehicle Engineering | Seoul, Korea
**In progress:** AI Mobility Systems

## Research Interests

Autonomous-driving perception and decision-making, with an emphasis on learning-based and end-to-end policies and their closed-loop behavior under occlusion and input delay.

## Research & Project Experience

### Kookmin Autonomous Driving Competition - AI-Edge Challenge

Apr 2026 - Aug 2026
**Team Lead** | Team of 5

- Built and integrated a ROS 2 steering pipeline spanning data collection, imitation-learning training, TorchScript inference, and switching between learned and rule-based control.
- Prepared and reviewed Roboflow datasets and trained a unified YOLO detector for traffic lights, vehicles, obstacles, and cones.
- Integrated mission-management and driving modules and tuned the competition vehicle. The team used rule-based control in the final competition after persistent oscillation in the learned steering policy.
- Defined parking waypoints on a SLAM map and implemented reverse recovery logic for the parking event.

Team results: **Preliminary 2nd Place** | Finals 13th Place (course completed) | **Parking Special Award**

### 2026 Student AI · SW Mobility Competition

Mar 2026 - Ongoing
Team of 27 | Decision-Making Subteam (5)

- Demonstrated real-vehicle waypoint driving at **10 m/s (36 km/h)** on both preliminary and final courses during venue practice on Sep 19-20, 2026.
- Developed ROS 2 cone-boundary paths, red-cone stop commands, and fail-safe logic. Following the February hybrid-control proposal, contributed Pure Pursuit–Stanley code and steering constraints; parameter tuning is ongoing.
- Compared hybrid tracking with Stanley alone in matched Gazebo segments at 6 m/s; reduced 95th-percentile heading error by approximately **14%** and steering-command total variation by approximately **10%**.
- Authored the high-level control section of Team K.A.I.'s competition technical report, documenting hybrid path tracking, steering constraints, and vehicle-command integration.
- Confirmed red-cone-triggered AEB stop commands in a 50 km/h vehicle trial; a brake hardware fault prevented an in-zone stop.

### Camera-Based V2I Cooperative End-to-End Autonomous Driving

2026 - Ongoing
Capstone Design Project | Team of 5 | Research & Design Phase

- Reviewed UniV2X's vehicle–infrastructure fusion framework and open-loop evaluation limitations, and prepared a literature summary to inform project scoping.
- Proposed a simplified design combining infrastructure object information with camera-based end-to-end planning. Contributed to scoping a subsequent closed-loop comparison on a small-scale vehicle; current work is limited to research and design.

### UAV Waypoint Mission - ROS 2 / PX4 SITL

May 2026 - Jun 2026
Autonomous System Platform Course | UAV Waypoint Development | Team of 4

- Developed a ROS 2 mission package with CSV waypoints, TF-based arrival checks, and coordinate-mode handling; integrated it with an existing PX4 offboard controller via PoseStamped commands.
- Demonstrated sequential waypoint tracking and final autonomous landing in the PX4 SITL/Gazebo course simulation.

### ROS 2 / CARLA Decision and Control

Mar 2026 - Jun 2026
Dream Semester Project | Decision-Making | Team of 3

- Implemented ROS 2 decision-making, Pure Pursuit waypoint tracking, and longitudinal control with CARLA integration and later vehicle-interface adaptation.
- Investigated simulated tracking instability by adjusting steering direction, lookahead, smoothing, and command limits; instability remained unresolved, with no real-vehicle run.

## Awards & Competitions

**2026 Kookmin AI-Edge Challenge**
Parking Special Award (team)

**2026 Hyundai Mobis Technical Special Award**
Autonomous Driving Division | Team K.A.I. (team award)
Feb 10, 2026 | Oral presentation, jointly credited to Woojin Hwang and Junghun Hwang: *Decision Logic: Combining Pure Pursuit and Stanley for Scenario-Adaptive Control*.

**2022 Drone Clash, General Division**
3rd Place

## Technical Skills

**Languages:** Python, C++

**Autonomous Systems:** ROS 2, CARLA, Gazebo, PX4 SITL

**AI / Computer Vision:** PyTorch, YOLO, OpenCV, Roboflow

**Tools / Environment:** Ubuntu, Git/GitHub, Foxglove, RViz2, rosbag2, QGroundControl